

Nalini Bisht

Houston, TX - 77004

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EDUCATION

- **Doctor of Philosophy (PhD) in Immunology and Microbiology | 2021 – Present**
Baylor College of Medicine
- **Master of Science (MS) in Medical Biotechnology (MBT) | 2016 – 2018** |
UIC College of Medicine at Rockford – 3.92/4.00
- **Bachelor of Engineering (BE) in Biotechnology | 2012 – 2016** |
Sri Jayachamarajendra College of Engineering, Mysuru, India – 9.38/10.00

PROFESSIONAL EXPERIENCE

Associate Scientist I, Biology

Enanta Pharmaceuticals, Inc.

May 2018 - May 2021

- Cell based assay development for the Coronavirus project
- Expansion and titring different coronavirus strains (BSL-2)
- Mutant virus rescue using the reverse genetics system
- Compound screening against strains A and B of the Respiratory syncytial virus (RSV)
- Generation, sequencing and characterization of RSV drug resistant mutants
- Expansion and titring of 20+ laboratory and clinical strains of RSV-A and B
- Exploring different cellular targets of RSV using RNA interference
- Testing effects of drug treatment on viral mRNA using Northern Blotting
- Laboratory supplies stock maintenance
- Data Presentation in Bi-weekly meetings

Graduate Teaching Assistant: Molecular Biology

UIC College of Medicine at Rockford

Aug 2017 – Dec 2017

- Organizing lab sessions with the course coordinator for masters of Medical Biotechnology students and helping them carry out the respective experiments; Proctoring exams and lab sessions
- Conducting lab math sessions for masters' students of Medical Biotechnology for learning laboratory math skills

Graduate Student Researcher

UIC College of Medicine at Rockford

Dec 2016 – Dec 2017

Immunomodulatory effects of r*Bma*Cystatin on Ulcerative Colitis and type-1 Diabetes

- Cloning and expression of recombinant *Brugia malayi* Cystatin
- Immunomodulation by r*Bma*Cystatin in RAW-264.7 cell lines
- Therapeutic potential of r*Bma*Cystatin in colitis and T1D mouse models

Student Researcher

Sri Jayachamarajendra College of Engineering (SJCE), India

Dec 2015 – May 2016

Biogenic Synthesis of Metallic Nanoparticles and their activity against Selective Pathogens

- Synthesis of nanoparticles from biogenic sources and their characterization
- Activity of synthesised nanoparticles against selective pathogens

Sri Jayachamarajendra College of Engineering, India

Oct – Nov 2015

Assessment of Anti-Bacterial activity of Citrus Fruits Peel Extracts on *Helicobacter pylori*

- Action of methanolic extracts of citrus fruit peels on *H. pylori*

Summer Research Fellow

Indian Institute of Science, India

May – July 2015

Role of TCP4 and Tarani on Leaf Epidermal cell differentiation in *Arabidopsis thaliana*

- Role of the TCP4 protein in controlling epidermal cell number and size
- Role of the *tarani* gene in trichome branching

PUBLICATIONS

- Michael H. J. Rhodin, Archie C. Reyes, Anand Balakrishnan, **Nalini Bisht**, Nicole M. Kelly, ..., Guoqiang Wang, Bryan Goodwin & Yat Sun Or

The small molecule inhibitor of SARS-CoV-2 3CLpro EDP-235 prevents viral replication and transmission in vivo. Nat Commun. 2024; 15: 6503

- Amana, Sharon Bright, Damilola Oyewole-Said, Keenan J. Ernste, **Nalini Bisht**, Arnav Murthy, Jonathan Vazquez-Perez, Vanaja Konduri, and William K. Decker.

The mARS complex: a critical mediator of immune regulation and homeostasis. Front Immunol 15 (2024): 1423510

- Kathryn M. Jones, Bin Zhan, Keenan Ernste, Maria Jose Villar, **Nalini Bisht**, Duc Nguyen, ..., Colby Hofferek, William K. Decker, Vanaja Konduri

Immunomodulatory proteins from hookworms reduce cardiac inflammation and modulate regulatory responses in a mouse model of chronic Trypanosoma cruzi infection. Front Parasitol. 2023; 2: 1244604

- **Nalini Bisht**, Vishal Khatri, Nikhil Chauhan and Ramaswamy Kalyanasundaram.

Cystatin from Filarial Parasites Suppress the Clinical Symptoms and Pathology of Experimentally Induced Colitis in Mice by Inducing T-Regulatory Cells, B1-Cells, and Alternatively Activated Macrophages. Biomedicines (2019). 7(4). 85.

- Baker Syed, **Nalini Bisht**, Prithvi S Bhat, Nikhil Karthik R, Ashwini Prasad., Dhananjaya, B. L., Satish, S., Hara Prasad, Nagendra Prasad, M. N.

Phytogenic synthesis of nanoparticles from Rhizophora mangle and their bactericidal potential with DNA damage activity. Nano-Structures & Nano-Objects (2017). 10:112-115

- Baker Syed, Prithvi S Bhat, **Nalini Bisht**, R. Nikhil Karthik R, Davronov Farangis-Rakhimova, Aishwarya T Devi, Ashwini Prasad, S. Satish, S. Nanjundaswamy & M.N. Nagendra Prasad.

Phytobiological Mediated Production of Silver Nanoparticles from Garcinia indica and Their Bactericidal Potential. Journal of Biologically Active Products from Nature (2018). 3:154-161

- Baker Syed, Nikhil Karthik, Prithvi Bhat, **Nalini Bisht**, Ashwini Prasad, S Satish, M.N. Nagendra Prasad.

- *Phyto-biologic bimetallic nanoparticles bearing antibacterial activity against human pathogens.*
Journal of King Saud University (2018) – Short communication

Phyto-biologic bimetallic nanoparticles bearing antibacterial activity against human pathogens.
Journal of King Saud University (2018) – Short communication

CONFERENCES AND POSTERS

- **Nalini Bisht**, Keenan J Ernste, Sharon Amany Bright, Harrison Berger, Sruthi Panja, Sujith K Joseph, William K Decker, Jill P Smith, Vanaja Konduri
“CCK-B Receptor Blockade with Proglumide Enhances Antitumor Activity and Persistence of CD8⁺CD161⁺ CAR-T Cells in Pancreatic Ductal Adenocarcinoma” – **AACR SC: Advances in Pancreatic Cancer Research**
Boston, MA, Sept 2025
- **Nalini Bisht**, Damilola Oyewole-Said, Keenan Ernste, Vanaja Konduri, William K Decker
“Dendritic cell CTLA-4⁺ extracellular vesicles as regulators of T cell polarization” – **IMMUNOLOGY2024TM, Annual meeting of AAI**
Chicago, IL, May 2024
- Anand Balakrishnan, Archie Reyes, Ruichao Shen, **Nalini Bisht**, Joyce Sweeney, Rachel Levene, Nicole McAllister, Tessa Cressey, Nathan Manalo, Michael H.J. Rhodin, Michael Vaine, Guoqiang Wang, Yat Sun Or, Bryan Goodwin
“Molecular basis for antiviral action of EDP-235: A potent and selective SARS-CoV-2 3CLpro inhibitor for the treatment of COVID-19” – **Experimental Biology 2022, Annual meeting of ASBMB**
Philadelphia, PA, April 2022
- **American Society for Virology (ASV), University of Minnesota Minneapolis, MN, July 2019**
- **Nalini Bisht**, Vishal Khatri, Nikhil Chauhan, Connie Vitali, Ramaswamy Kalyanasundaram.
“*Brugia malayi* recombinant cystatin ameliorates development of murine colitis by induction of IL-10⁺ T-regulatory cells and IL-10⁺ B1-cells” – **Federation of Clinical Immunology Societies (FOCiS)**
San Francisco, CA, June 2018
- **Federation of Clinical Immunology Societies (FOCiS)**
Chicago, IL, June 2017
Baker. S., Karthik. R. N., Dhananjaya. B. L., **Bisht. N.**, Bhat. P. S., Shejwadkar. B. N., Bhandarkar. S.S., Srinath. L.N.V.M., Monisha. M., Satish. S., Nagendra Prasad. M. N.
“Phytobiologic Synthesis of Silver Nanobactericides using *Annona squamosa* and their activity against test pathogens with DNA Damage Property” – **Functional Nanomaterials for Energy, Environment and Health**
Mangalore, India, March 2016
- Baker. S., **Bisht. N.**, Bhat. P. S., Karthik. R. N., Shejwadkar. B. N., Bhandarkar. S.S., Srinath. L.N.V.M., Monisha. M., Satish. S., Nagendra Prasad. M. N.
“Phytobiologic Synthesis of Silver Nanobactericides using *Rhizophora mangle* L. and their activity against test pathogens with DNA Damage Property” – **Application of Modern Analytical Techniques to Fundamental Research in Chemistry**
Mysuru, India, Feb 2016
- Ramu. R., Shirahatti. P.S., Bhat. P.S., **Bisht. N.**, Nagendra Prasad. M. N.

“Impact of Active Compounds isolated from Banana flower and pseudostem towards cytoprotective and DNA protection activities” – **Phytochemicals and Functional Foods: Current Situation and Future Prospects** **Mysuru, India, Feb 2015**

- Shirahatti. P.S., **Bisht. N.**, Bhat. P.S., Nagendra Prasad. M. N.
“Accurate identification and in vitro control of *Phomopsis azadirachtae*, the causative organism of die back disease of Neem” – **Microbes and Human Welfare** **Mysuru, India, March 2015**

AWARDS AND ACHIEVEMENTS

- **Best Poster presentation** at the 6th annual Gulf Coast Consortia Future of Immunology Symposium **Houston, TX, June 2026**
- **First place in Best Oral presentation** at the 2026 Trainee Research symposium organized by the Dept of Pathology and Immunology at Baylor College of Medicine **Houston, TX, Feb 2026**
- **First place in Best Oral presentation** at the 2025 Trainee Research symposium organized by the Dept of Pathology and Immunology at Baylor College of Medicine **Houston, TX, April 2025**
- **Third place in Poster presentation** at the 5th annual Gulf Coast Consortia Future of Immunology Symposium **Houston, TX, April 2025**
- **Second place in Lightning poster talk presentation** at the 5th annual Gulf Coast Consortia Future of Immunology Symposium **Houston, TX, April 2025**
- **Second place in the Elevator pitch competition** at the Graduate Student Symposium organized by Baylor College of Medicine **Houston, TX, May 2024**
- **Outstanding Research Award** at the 2018 Graduation Ceremony organized by the UIC College of Medicine, Rockford **Rockford, IL, Dec 2018**
- **Best Basic Science Research Poster Award** at Research Day organized by the UIC College of Medicine, Rockford **Rockford, IL, April 2018**
- **Board of Trustees Tuition waiver award** for the Fall 2016 semester by the Department of Biomedical Sciences at UIC College of Medicine, Rockford. **Rockford, IL, Fall 2016**
- **Best poster award** at a UGC sponsored Two-day National Conference on Application of Modern Analytical Techniques to Fundamental Research in Chemistry organized by JSS College of Arts, Commerce and Science, India **Mysuru, India, Feb 2016**